William Hunter McNichols (he/him)

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Overview

In my research, I study how large language models (LLMs) are reshaping education by exploring the capabilities of these tools to create new learning experiences and measuring their impact in authentic learning environments. I employ post-training techniques such as supervised fine-tuning (SFT) and preference alignment (e.g. DPO) to align LLMs to educational tasks, enabling applications in personalized content generation, student modeling, and tutoring dialogue systems. Furthermore, I design and implement classroom studies to analyze the real-world impact that AI has on students' learning outcomes. Beyond research, I previously worked as a software engineer building distributed systems for Siri at Apple and taught university AI courses. These experiences enable me to translate research findings into production systems and vice versa.

Education

University of Massachusetts Amherst — Amherst, MA
Ph.D. in Computer Science (Expected September 2026) — *GPA: 3.95*

University of Southern California — Los Angeles, CA
M.S. Computer Science, 2016 — *GPA: 3.64*

University of Southern California — Los Angeles, CA
B.S. Computer Science, 2016 — *GPA: 3.73, Magna Cum Laude*

Experience

PhD Research: AI for Education — University of Massachusetts Amherst : Sep 2022 – Present

- Develop large language model systems for personalized tutoring and learning applications, focusing on post-training methods including fine-tuning, preference alignment, and retrieval-augmented prompting.
- Built and deployed an LLM-powered tutoring platform used by 300+ students in an AI course, collecting 16K+ student–model interactions to study user behavior and model adaptation in real-world settings.
- Designed evaluation frameworks combining human annotations, behavioral analysis, and automated metrics to assess model feedback quality, reasoning, and pedagogical alignment.
- Developed LLM pipelines for student modeling and content generation, including automated feedback generation, error classification, and distractor generation for math questions.
- Published research at NLP and AI venues (EMNLP, NAACL, AIED, EDM, LAK) on LLM evaluation, personalization, and educational applications.

Applied Scientist, Software Inc. (Acquired by OpenAI) — San Francisco: May 2025 – August 2025

- Investigated "Computer Use" LLM capabilities of open source LLMs integrated into an interactive desktop application, combining research and software engineering skills.
- Built an LLM training and deployment infrastructure within a small company's software ecosystem.
- Mentored a junior research engineer on a distributed evaluation project, teaching principles of software design, experimental methodology, and applied machine learning.
- Demonstrated leadership through collaborative planning, project management, and the translation of research goals into mentorship opportunities.

Software Engineer, Siri Distributed Platform, Apple Inc. — Cupertino, CA : 2016 – 2019

- Developed and maintained internal frameworks to support scalable Siri dialogue systems.
- Worked across platforms using Java, TypeScript, Python, and Objective-C
- Created internal tools and extensions for VSCode, significantly simplifying the developer experience.
- Modernized JavaScript runtime integration within Objective-C systems.
- Automated developer-facing workflows in the Siri personality pipeline, increasing throughput by over 500%.

Founder & Freelance Software Engineer, HunterCodes — Remote : 2019 – 2022

- Founded and operated an independent software development business
- Led all aspects of product development from client discovery to implementation and deployment.
- Partnered with startups and small businesses to design and deploy full-stack web applications and data pipelines.
- Delivered custom, full-stack software solutions across a wide range of frameworks such as React and AWS

Skills

Machine Learning: PyTorch, HuggingFace, vLLM, Deep Learning, LLM fine-tuning and Alignment, NLP Evaluation

Product & Infrastructure: AWS, Linux, Git, Docker, React, SLURM

Coding Languages: Python, JavaScript, Java, Objective-C, C++, TypeScript

Publications

McNichols, H., Ikram, F., and Lan, A. (2026). *The StudyChats Dataset: Analyzing Student Dialogues with ChatGPT in an Artificial Intelligence Course.* The 16th International Analytics and Knowledge Conference

Tran, P., Feng, W., Sireci, S. G., **McNichols, H.**, and Lan, A. *Using Item Features to Calibrate Educational Test Items: Comparing Artificial Intelligence and Classical Approaches. American Journal of Education and Learning*, 10(2), 178–189, 2025.

Lee, J., **McNichols, H.**, and Lan, A. *Exploring Automated Keyword Mnemonics Generation with Large Language Models via Overgenerate-and-Rank.* Findings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2024.

Feng, W.*, Lee, J.*, **McNichols, H.***, Scarlatos, A.*, Smith, D., Woodhead, S., Ornelas, N., and Lan, A. *Exploring Automated Distractor Generation for Math Multiple-Choice Questions via Large Language Models.* Findings of the Association for Computational Linguistics: NAACL, 2024.

McNichols, H., Lee, J., Fancsali, S., Ritter, S., and Lan, A. *Can Large Language Models Replicate ITS Feedback on Open-Ended Math Questions?* Proceedings of the 17th International Conference on Educational Data Mining (EDM), 2024.

McNichols, H.*, Feng, W.*, Lee, J.*, Scarlatos, A., Smith, D., Woodhead, S., and Lan, A. *Automated Distractor and Feedback Generation for Math Multiple-choice Questions via In-context Learning.* GAIED Workshop at NeurIPS, 2023.

Kumar, N. A., Feng, W., Lee, J., **McNichols, H.**, Ghosh, A., and Lan, A. *A Conceptual Model for End-to-End Causal Discovery in Knowledge Tracing.* Proceedings of the 16th International Conference on Educational Data Mining (EDM), 2023.

McNichols, H., Zhang, M., and Lan, A. *Algebra Error Classification with Large Language Models.* International Conference on Artificial Intelligence in Education (AIED), 2023.

*Denotes equal contribution.

Teaching Experience

University of Massachusetts Amherst Amherst, MA : 2022 – Present

- **CS 383 Artificial Intelligence (Fall 2024, Spring 2025) — Instructor**
  - Taught classrooms of 180 undergraduate students, scaling a curriculum developed for smaller classrooms.
  - Conducted classroom research evaluating student interactions with GPT-powered Intelligent Tutoring Systems.
  - Managed a course staff of 12 graders and TAs; coordinated grading, feedback, and classroom logistics.
  - Developed coursework, including new assignments and assessments, designed for independent student learning.
- **CS 240 Reasoning Under Uncertainty (Fall 2025) — Teaching Assistant**
- **CS 326 Web Programming (Fall 2022) — Teaching Assistant**

City College of New York Manhattan, NY : Jan 2020 – Sep 2022

- **CS 44800 Artificial Intelligence — Lecturer**
  - Designed and taught an advanced undergraduate elective to roughly 30 students, emphasizing foundational AI concepts and recent industry trends.
  - Created hands-on coding assignments in Python, including Support Vector Machines, Neural Networks, and other machine learning algorithms.
  - Adapted coursework regularly to incorporate emerging technologies and reflect the latest industrial applications.

Honors and Awards

NAEP Math Scoring Challenge (2023) – Grand Prize Winner

NeurIPS 2022 Causal Edu Competition - 3rd place

University of Southern California Trustee Scholar